

Element L

We could've read the materials list more in depth because it set us back. Our team was forced to redesign the box to store the plant due to wood that was either bowed or just plain weak. The wood we initially used was so weak that when we drilled in screws, it would split into layers. So if we read it correctly the first time we could have possibly programmed it to be automatic. After the blinds were put onto the prototype, we were done. But, me and one of my team members came up with the idea of an automatic water sprinkler system for the plant. Unfortunately, we were unable to implement this idea, because of the waiting time for the materials to arrive and the design-a-thon day was only a week away. But, if our team had just a little more time, we would have set the water-sprinkler system up, so that the indoor plant can have sunlight and water, both necessary to carry out a plant's daily functions, and for the plant to thrive.

Our team would keep the 3D printed clips because they were very helpful and we wouldn't have been able to finish the design without them. Even though they may look like it doesn't contribute much to the project; without the clips, our blinds wouldn't have stayed on top of the box. Honestly, the process of making the 3-D clips was the longest part, due to the time to create it on onShape, and the time it took to print it out in class, which took off 2 class periods. All in all, we would keep the 3-D clips, because it turned out to be the most useful part of our project, and if our team kept them, it would save us so much more time, more time to think about how to better improve our prototype.

Our team would most definitely change the blinds. Not exactly changing it, but changing the material of the blinds. Our current blinds are weak, and when each individual blind is held back against the 3-D printed clips for tension, when released it easily becomes wrinkled and is not back in its original shape. For this project, our team really wasn't forced onto a budget, and we also didn't really spend much, other than the blinds. All other parts of the project were either from home, or it was made in class, like the 3-D clips. Our team thought that, if we spent money on quality-blinds rather than just cheap material, not only would our blinds look better, it would've saved us more time. With the blinds we used at first, it was so fragile, that at last we had to get rubber bands to hold back a couple of individual blinds that were just hanging down.

Our actual work-time for this project was in the months of January and February, with the design-a-thon being at the end of February. Unfortunately, this year our school had many snow-days, and were on remote-learning days. This prevented us from working on our project, because it had to stay at school. Along with this, easy accessibility to materials would've helped us to complete the project more efficiently and quicker. Our team spent 2 class periods, just waiting for our materials to come in. We had to redesign our box 2 times because the wood was not strong enough to withstand the force of screws being drilled in. These were probably the most problems that could've been solved in order for our team to complete the project easier and faster.